

Effort Across Stochastic, Controlled, and Adaptive Worlds

Epistemic License, Changed Agents, Interaction Separation, Learning-Bearing Retry, and Rational Exit

An attack-first readout architecture for deciding when randomness is licensed, when local theorems may travel, and when another try ceases to be the same experiment

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Version note (v0.4: Uplift edition — changes from v0.3): This edition applies the six items raised by the independent glosa review (`ASSESSMENT_2026-09-07_effort_v0_3_glosa.md`, verdict PASS with warnings). (1) Appendix D is renamed "Scope of Contribution" and its opening sentence — which disclaimed authorship of the imported external mathematics — is reworded to avoid the rule-6 priority word it previously used, without changing what it disclaims or claims. (2) Section 1's "This yields the first non-collapse of the revised theory" is reworded to "This yields the initial non-collapse of the revised theory," removing an ordinal use of a rule-6 word. (3) References [1]–[8] are pinned: [1] Toledo now carries the concept DOI 10.5281/zenodo.22537318 and the current release DOI (Version 1.4.0) 10.5281/zenodo.22637913; [2] Readout Genesis and [3] Readout Universe are pinned to the public commit anchors recorded in Toledo's own import provenance (`coq/readout_genesis/PROVENANCE.json`, `coq/readout_universe/PROVENANCE.json`); [4] Shared-Readout Non-Identifiability, [6] SkillMe, [7] Finite Readout Acceleration, and [8] Retained Multilevel Sturm are pinned to their current public HEAD commits; [5] Zero-Readout Certifies is pinned to its own Zenodo DOI 10.5281/zenodo.21665100. (4) A DOI or ISBN is added to the three of the four previously bare monographs whose edition could be independently confirmed against the reference line as given ([9] Doob 1953, ISBN 0-471-52369-0; [15] Pearl 2009 2nd ed., ISBN 978-0-521-89560-6; [17] Husserl 1991 Kluwer, ISBN 0-7923-0891-3); [18] Merleau-Ponty 2012 Routledge trans. Landes carries no confirmable DOI or ISBN from this reference line alone and is marked "no DOI; publisher edition as listed." (5) Appendix A's provenance ledger gains the missing Eq. 4 row (reader-preservation clause, `weld/M.02.v1`), closing the gap between Eq. 3 and Eq. 5 the review flagged. (6) A K-State / Claim Boundary (K0) block and a Core Epistemic Registration block (glosa P20 template) are added after the byline. No Markdown or LaTeX source of v0.3 (or v0.2/v0.1) exists anywhere on this machine; this Markdown was reconstructed from `pdftotext -layout` of the deposited v0.3 PDF plus direct visual inspection of every page for equation fidelity (all 41 numbered equations and both multi-part tables were checked against the rendered PDF page images, not the raw column-merged text alone). No sentence, equation, reference, or provenance tag from v0.3 was altered beyond the six items above; every other word is carried over unchanged.

Core Epistemic Registration

This manuscript uses the three-part project registration protocol used elsewhere in the Human-AI Readout Programme (glosa P20 template). Registration is an attribution device, not a certification of truth or expertise.

Core Respondent / Experience-Based Expert: Yaoharee Lahtee — lived experience in independent, cross-domain theory construction under the Human-AI Readout Programme; selections: the attack-first ordering, the four v0.3 closures (conditional/strong stochastic license separation, witness-preserving cross-domain objection, positive interaction-separation condition, local-theorem containment), and every definition, derivation, proposition, and non-collapse in this manuscript; interpretation: all claims and their stated scope are the author's own.

Interactional Expert: None

AI Model(s) Used: The v0.3 manuscript's own text names no AI model or vendor anywhere in its body, abstract, ledger, or references — confirmed by the independent glosa review's rule-9 scan (`ASSESSMENT_2026-09-07_effort_v0_3_glosa.md`, zero matches for any AI/vendor term). No model is retroactively attributed here for that draft; its production history beyond the deposited text is not disclosed and is not asserted.

AI Model(s) Used (v0.4 uplift edition): Claude Fable 5.1 (via Claude Code) — independent glosa review (rules 6/9/17, tier honesty, and Toledo-citation/commit-anchor verification), reconstruction of this manuscript's Markdown source from the deposited v0.3 PDF, DOI/ISBN verification for the reference list, completion of the Appendix A provenance ledger, and PDF typesetting for this v0.4 uplift edition.

Non-collapse rule:

Core experiential source $\neq$ Interactional expert $\neq$ AI model.

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K-State / Claim Boundary (K0)

This manuscript's own registration tier, in the discipline used elsewhere in the Human-AI Readout Programme, is **K0**: every equation newly introduced here (Eqs. 1, 6, 9–41) is asserted as a structural definition, derivation, proposition, or synthesis, not as a machine-checked (`Th_coqc`) or empirically-diagnosed (`finite_diagnostic`) result. Appendix A already tags each new item this way; this block states the boundary once, in one place, rather than leaving it implicit across forty individual footnotes.

What is claimed: that the licensing and typing architecture in Sections 1–18 is internally consistent given its own stated definitions; that the five equations imported from Toledo (Eqs. 2, 3, 4, 5, 7, 8) are cited and glossed faithfully to the current `morrocwi/toledo` registry, as verified in Appendix A; and that the architecture absorbs, without contradiction, the external results it names (Doob; Gittins; Lai and Robbins; Kaelbling,

Littman, and Cassandra; Shapley; Aubin, Bayen, and Saint-Pierre; Pearl; Rudolph and Ross; Husserl; Merleau-Ponty; Halák; Stone and Bruineberg) under the licensing conditions Sections 4, 5, and 11 state for each.

What is not claimed: that any of Eqs. 1, 6, 9–41 has been machine-checked, empirically tested, or independently peer-reviewed; that the thresholds τ_I , τ_E , τ_G , α , ε_C named throughout carry any declared numeric value — each is a placeholder for a domain-specific preregistration, per Section 17; that this manuscript's own new material outranks, supersedes, or requires the retraction of any external theorem it discusses (Section 15 and Appendix B state explicitly what each rival formalism still supplies on its own); or that this architecture has been applied to a completed empirical case (Section 17 is a preregistration programme for future work, not a finished study). No external certifying authority is sought for this internal architecture; the standard applied is whether the stated definitions cohere and survive their own declared falsifiers (Eqs. 26–27 and 40, and the falsifier list in Section 17).

Anchor and revision status. This manuscript keeps *Effort Under Stochastic and Other-Controlled Worlds* v0.1 as its anchor and v0.2 as its immediate mathematical scaffold. Version 0.3 is an adversarial elevation, not a replacement. It retains event/control typing, source identity status, changed-agent interaction, revision/learning separation, typed cost-risk viability, and regime-specific routing. It adds four closures forced by hostile review: (i) a distinction between conditional stochastic modeling and strong stochastic attribution; (ii) a witness-preserving cross-domain objection license, so enlarging a filtration or state vector does not by itself prove epistemic or phenomenological sufficiency; (iii) a positive interaction-separation condition with an explicit falsifier, so changed agents receive no causal credit unless the encounter is demonstrably sensitive to the retained change; and (iv) a local-theorem containment principle preventing a theorem that is correct in one formalism from silently universalizing beyond the distinctions its bridge preserves. Mathematical provenance follows the Toledo source policy. Existing equations are only those verified in `morrocwi/toledo`; every new equation remains explicitly labeled. Ontology is interpreted through Readout Genesis and epistemology through Readout Universe.

Elevated central claim. A theory of retry is incomplete merely because it specifies an outcome distribution. It must state whether stochasticity is assumed, supported, or strongly attributed; what the focal agent can read; what is retained; whether the retained difference becomes certified learning; whether the effective agent changes in a distinction to which the next interaction is actually sensitive; who else controls the outcome; and whether another attempt remains viable under finite cost and risk. A local theorem may fully defeat a local claim — for example, optional stopping may certify a null-effort payoff regime — but it earns no automatic right to erase epistemic, phenomenological, or agent-world transformation that its state/readout bridge does not preserve.

Abstract

Theories of effort are often criticized by appeal to randomness: if outcomes are stochastic, persistence may merely expose an agent to more draws without improving expected value, and optional-stopping results can block naive claims that "trying longer" creates an edge in a fair process. This criticism is valid inside its mathematical domain but is often granted more trans-domain authority than its representation has earned. The label *random* may denote an ontic generative mechanism, a conditional stochastic model, an epistemic predictive distribution under incomplete access, a hidden decision by another agent, or an unresolved residual. Moreover, even when a world-side source is unchanged to the strongest resolution a finite reader can certify, retained experience can alter the effective reader, policy, skill, embodied disposition, or live accessibility with which the next event is encountered.

Version 0.3 elevates an earlier effort-under-uncertainty architecture in four ways. First, it separates a *Conditional Stochastic Modeling License*, which permits theorem use under declared assumptions, from a *Strong Stochastic Attribution License*, which permits stochasticity itself to be treated as a load-bearing property of the event. Second, it strengthens the Cross-Domain Objection License by requiring preservation of all load-bearing witnesses, not only a transition and output variable. Third, it introduces a positive *Interaction-Separation Condition*: a changed agent receives explanatory credit only when a preregistered task reader detects a material change in the subsequent agent-world interaction under a held or robustly equivalent source. Fourth, it contains local theorems to the distinctions their bridges preserve, while routing licensed regimes to martingale/null-process analysis, controlled stochastic or bandit models, partially observable models, stochastic games, or mixed/unresolved treatment. Retry is separated into exposure, retained revision, certified learning, improvement, and action-bearing repetition without new learning. A typed risk/viability layer then governs continuation and exit. The resulting thesis is stronger than "effort sometimes changes odds": before another try can be called the same experiment, the source, effective agent, interaction, control locus, learning status, and admissible mathematical representation must all survive the declared readout.

1. Why Randomness Is Not a Self-Licensing Objection

Suppose an agent applies repeatedly for a job, grant, publication, visa, treatment, admission, collaboration, or institutional permission. The eleventh attempt succeeds after ten failures. The observed sequence is compatible with many mechanisms: the agent learned; the agent merely received another independent opportunity; a committee changed; a hidden threshold moved; another actor made a different choice; the environment adapted strategically; or a favorable stochastic realization occurred. A single success cannot identify among these explanations.

The first draft of this theory therefore distinguished exogenous stochasticity, skill-sensitive stochasticity, other-agent control, coupled adaptation, and mixed processes. The

present revision pushes one level deeper. Before asking which stochastic theorem applies, one must ask whether *stochasticity itself has been warranted as the relevant description*.

Attack-first question. *When a critic says "the outcome is random," what exactly has been established? Has a world-side stochastic mechanism been identified? Has only a predictive distribution been fitted? Is another agent's hidden decision being represented as noise? Or has an unresolved remainder simply been assigned a random variable? If these possibilities have not been separated, the stochastic objection is conditional at best.*

This yields the initial non-collapse of the revised theory:

$$\begin{aligned} \text{uncertainty} &\neq \text{stochastic mechanism} \\ \text{stochastic mechanism} &\neq \text{other-agent control} \\ \text{other-agent control} &\neq \text{unresolved residual} \end{aligned} \tag{1}$$

[NEW NON-COLLAPSE: rooted mathematically in Toledo domain/readout separation and interpreted epistemically through Readout Universe; connects four distinct sources of an apparently unpredictable outcome so that ignorance is not silently promoted into ontology.]

The claim is not that stochastic mathematics is weak. It is that its domain of application must itself be read out and licensed. Optional stopping, bandit theory, POMDPs, and stochastic games become powerful once the event has been correctly typed; none of them, by theorem alone, identifies which type of event one is facing.

2. Anchor: The Toledo Dynamics Retained from v0.1

The mathematical anchor remains Toledo's state stepper:

$$S_{n+1} = F(S_n, u_n, c_n, T_n) \tag{2}$$

[TOLEDO EXISTING: `weld/M.01.v1`; current Toledo registry records the state stepper as part of the root weld. It connects retained state S_n , control/input u_n , context c_n , and ordered history T_n to the next state.]

A domain interpretation is admissible only when the translation preserves the dynamics:

$$q_D(F(z, u, c, T)) = F_D(q_D(z), u, c, T) \tag{3}$$

[TOLEDO EXISTING: `weld/M.02.v1`; Toledo admissibility condition connecting root dynamics to a declared domain dynamics.]

The associated reader must also be preserved:

$$O_D(z; Q, c) = O_D^\#(q_D(z); Q, c) \tag{4}$$

[TOLEDO EXISTING: `weld/M.02.v1`; reader-preservation clause.]

Toledo already supplies a finite-horizon reader-equivalence relation:

$$z \sim_{Q, O, c, L} z' \iff O(F^k z) = O(F^k z') \quad \forall k \leq L \tag{5}$$

[TOLEDO EXISTING: `weld/M.03.v1`; connects finite observational equivalence to a declared reader, query/context, and horizon.]

These equations are decisive for the present revision. They imply that a theorem imported from another domain is not automatically entitled to constrain the effort domain. The bridge itself must preserve the load-bearing dynamics and readout.

2.1 External encounter count is not internal update count

The original intuition also survives unchanged. Finite external opportunity does not force a one-to-one count of internal revision:

$$N_{ext} < \infty \not\Rightarrow N_{int} = N_{ext} \quad (6)$$

[NEW PROPOSITION: rooted in repeated admissible application of Eq. 2; connects a finite number of world-side encounters to potentially multiple internal state revisions without claiming infinite energy, infinite benefit, or monotonic improvement.]

The claim is structural rather than motivational. One external event can be re-read, practiced, modeled, rehearsed, reframed, or incorporated more than once. Whether such revisions are useful is a separate question.

3. Internal Epistemic Order: World Event Is Not Retained Directly

The strongest phenomenological criticism of v0.1 was correct: the paper moved too quickly from world feedback to retention. Toledo's current registry already contains a constitutional ordering rule:

$$\begin{aligned} &\text{retention} \rightarrow \text{structure} \rightarrow \text{translation} \rightarrow \text{readout} \\ &\rightarrow \text{meaning} \rightarrow \text{experience} \rightarrow \text{memory} \\ &\rightarrow \text{belief} \rightarrow \text{claim} \rightarrow \text{checking} \rightarrow \text{status} \end{aligned} \quad (7)$$

[TOLEDO EXISTING: `EQ-015/M.03.v1`; constitutional ordering rule in the current Toledo registry.]

This ordering prevents a world event from being treated as though it were copied directly into an agent. A result is accessed through a reader, acquires domain meaning, appears in experience, and only then can contribute to later memory, belief, or action.

The programme separately states:

$$\text{Exposure} \neq \text{Retention} \neq \text{Improvement} \quad (8)$$

[TOLEDO EXISTING: `A.5/H.09.v1`; non-collapse registered in Toledo.]

The phenomenological literature supplies an independent reason not to collapse encounters into isolated draws. Husserl's account of temporal consciousness treats present experience as structured by retention of the just-past and protention toward what is about to occur, rather than as a succession of independent "now" points. Merleau-Pontian accounts of habit and motor intentionality further emphasize that practical

understanding is embodied and historically sedimented rather than exhausted by propositional belief [17, 18, 19, 20].

Phenomenological compatibility claim. *A repeated outcome can be statistically generated by the same world-side mechanism while no longer being lived as the same encounter. The later agent may possess different anticipation, salience, bodily readiness, fear, familiarity, conceptual organization, or perceived action-space. A probability model of outcomes does not become a model of this transformation merely by assigning a random variable to the outcome.*

4. Two Licenses for Stochasticity: Modeling Is Not Attribution

Hostile review exposed an important overreach in v0.2: a strong criterion for calling an event stochastic should not be mistaken for a prohibition on conditional stochastic modeling. The present revision therefore separates two licenses.

Let r_n denote the finite record available under a declared source-reader O_Q^G . Define the admissible mechanism fibre:

$$\mathfrak{M}_Q(r_n) := \{m \in \mathfrak{M}_Q^{adm} : O_Q^G(m) = r_n\} \quad (9)$$

[NEW DOMAIN DEFINITION: rooted in Toledo reader/domain admissibility and finite reader-equivalence; connects an observed record to the set of mechanisms still compatible with that record instead of assuming a unique hidden cause.]

The admissible class is not the set of every logically imaginable story. It is the class that survives the declared evidence, background constraints, scope, and falsifiers. This prevents the HOLD state from degenerating into radical skepticism.

A stochastic model M may be used conditionally when its assumptions and scope are declared:

$$CSML_Q(M) = \text{PASS} \iff M \in \mathfrak{M}_Q^{adm} \wedge \text{Asm}(M, Q) \text{ is explicit} \quad (10)$$

[NEW DOMAIN DEFINITION: Conditional Stochastic Modeling License; rooted in the admissible model class after Toledo inspection. It connects a declared stochastic assumption to legitimate conditional theorem use without asserting that the world-side mechanism has been identified as stochastic.]

Thus a critic may validly prove: "if this payoff process is a martingale under filtration $\mathcal{F}_n$, then ..." whenever Eq. 10 passes.

A stronger claim — that stochasticity itself is the load-bearing mechanism at the declared resolution — requires stronger evidence. Partition the admissible class into stochastic and non-stochastic/other-controlled candidates. Define:

$$SAL_Q^{strong}(n) = \text{PASS} \iff \mathfrak{M}_Q(r_n) \subseteq \mathfrak{M}_Q^{stoch} \quad (11)$$

[NEW DOMAIN DEFINITION: Strong Stochastic Attribution License; rooted in Eq. 9. It connects finite evidence to permission to treat stochasticity as a property of the load-bearing mechanism rather than only a declared modeling assumption.]

If both stochastic and non-stochastic/other-controlled mechanisms remain admissible,

$$\begin{aligned} \mathfrak{M}_Q(r_n) \cap \mathfrak{M}_Q^{stoch} &\neq \emptyset, \\ \mathfrak{M}_Q(r_n) \cap \mathfrak{M}_Q^{nonstoch} &\neq \emptyset \\ \Rightarrow SAL_Q^{strong}(n) &= \text{HOLD} \end{aligned} \quad (12)$$

[NEW DERIVATION: rooted in Eq. 11; connects mechanism non-identifiability to an explicit HOLD state rather than falsely deciding "random" or "not random."]

The two licenses must not collapse:

$$CSML_Q(M) = \text{PASS} \not\Rightarrow SAL_Q^{strong}(n) = \text{PASS} \quad (13)$$

[NEW NON-COLLAPSE: rooted in Eqs. 10–11; separates "a stochastic model is admissible under declared assumptions" from "the event has been strongly identified as stochastic."]

No Unlicensed Randomness Principle, strengthened. Conditional stochastic mathematics is always welcome when its assumptions are declared. What requires additional warrant is the move from a useful stochastic representation to a claim that randomness is the event's load-bearing ontology or to a trans-domain refutation of agent transformation. The architecture therefore does not block probability theory; it blocks ontology by notation.

5. Cross-Domain Objection License: A Larger Filtration Is Not Yet a Larger Explanation

Let $\mathcal{R}$ be an external formalism with state space Z_R , transition F_R , and readout O_R . Let D be the effort/agent-world domain with F_D and O_D . Version 0.2 required transition and output preservation. Hostile review adds a stronger requirement: if the objection is meant to erase an epistemic, phenomenological, skill, accessibility, or cost claim, the bridge must preserve the load-bearing witness for that claim as well.

Let $\mathbf{W}_Q^D = (W_{Q,1}^D, \dots, W_{Q,k}^D)$ be the finite family of declared load-bearing witnesses in domain D . A witness may measure, for example, calibration, embodied task capability, live accessibility, salience, policy repertoire, or a declared phenomenological contrast. A cross-domain bridge is licensed only if there exists a translation $T_{R \rightarrow D}$ such that

$$\begin{aligned} T_{R \rightarrow D} \circ F_R &= F_D \circ T_{R \rightarrow D}, & O_D \circ T_{R \rightarrow D} &= O_R, \\ W_{Q,j}^D \circ T_{R \rightarrow D} &= W_{Q,j}^R \quad \forall j \in J_Q. \end{aligned} \quad (14)$$

[NEW DERIVATION: witness-preserving Cross-Domain Objection bridge; directly patterned on Toledo `weld/M.02.v1`. It connects a theorem's native state space to the effort domain and requires preservation of every distinction the theorem is being used to constrain.]

Define:

$$CDOL_Q(\mathcal{R} \rightarrow D) = \text{PASS} \iff \exists T_{\mathcal{R} \rightarrow D} : (14) \text{ holds} \quad (15)$$

[NEW DOMAIN DEFINITION: Witness-Preserving Cross-Domain Objection License; rooted in Toledo domain-weld admissibility.]

This yields a three-part sufficiency burden whenever an external theorem is used against an internal transformation claim:

Cross-domain sufficiency triad.

1. **Dynamic closure:** the translated dynamics commute.
2. **Epistemic sufficiency:** the representation preserves the evidence/warrant distinction being challenged.
3. **Task-phenomenological sufficiency:** any lived, embodied, accessibility, or practical distinction that is load-bearing for the claim is either preserved by a declared witness or explicitly excluded from scope.

This directly answers the "enlarge the filtration" objection. A critic may put more history into $\mathcal{F}_n$; that produces a richer stochastic representation. It does not, by notation alone, prove that the filtration is sufficient for every task-relevant internal distinction. Sufficiency is earned by Eq. 14, not by the size of the sigma-algebra.

Local-Theorem Containment Principle. *A theorem that is correct in formalism $\mathcal{R}$ may constrain every claim whose load-bearing witnesses are preserved by an admissible bridge into D . It has no automatic force over distinctions erased by that bridge. Local victory is fully retained; silent universalization is not.*

If a martingale representation preserves the payoff, agent transformation, and every witness relevant to the disputed claim, optional-stopping results bite at full force. If it preserves only the payoff, it may defeat a payoff-edge claim while leaving a changed-agent claim untouched. This is not immunity from mathematics; it is typed scope.

6. Event-Control Typing With a Legitimate HOLD State

The v0.1 event taxonomy is retained but made explicitly epistemic. For a declared event e_n :

$$ECT_Q(e_n) \in \{E_0, E_1, E_2, E_3, E_4, \text{HOLD}\} \quad (16)$$

[NEW DOMAIN DEFINITION: rooted in Toledo domain typing and interpreted through Readout Universe's claim-boundary discipline; connects finite evidence to an event-control regime without forcing classification when the readout cannot support one.]

Event-control regimes.

- E_0 - **Exogenous stochastic/null-effort regime**. Focal control does not alter the load-bearing outcome law at the declared resolution.
- E_1 - **Agent-sensitive stochastic regime**. Outcomes remain stochastic, but focal action/skill/policy changes a relevant transition or reward law.
- E_2 - **Other-agent-controlled regime**. Another agent/institution selects a load-bearing component; uncertainty to the focal agent does not itself establish randomness.
- E_3 - **Coupled adaptive regime**. Focal and external agents both update in response to interaction.
- E_4 - **Identified mixed regime**. Multiple mechanisms are jointly evidenced and represented.
- **HOLD - Unresolved regime**. Available readout is insufficient to license a stronger attribution.

7. Source Identity Is Not Trial Identity

Version 0.1 defined "same random source" too weakly by equal marginal distributions. Equal marginals can coexist with radically different dependence structures and transition mechanisms. The corrected version uses Toledo's finite reader-equivalence.

Let $\mathcal{G}_n$ be the world-side source state/mechanism relevant to the event. We say only that two source states are reader-equivalent when

$$\mathcal{G}_n \sim_{Q,O,c,L}^G \mathcal{G}_m \quad (17)$$

[TOLEDO APPLICATION: application of `weld/M.03.v1` to the declared source domain; this is not a claim of latent mechanism identity.]

A separate status records the strength of the source comparison:

$$SID_Q(n, m) \in \{\text{SAME}_{\text{mech}}, \text{EQUIV}_Q, \text{DIFF}_Q, \text{HOLD}\} \quad (18)$$

[NEW DOMAIN DEFINITION: rooted in Toledo finite reader-equivalence; connects direct mechanism evidence, finite readout equivalence, distinguishability, and unresolved comparison without collapsing them.]

The critical distinction is therefore:

$$SID_Q(n, m) = \text{EQUIV}_Q \not\Rightarrow \mathcal{G}_n = \mathcal{G}_m \quad (19)$$

[NEW PROPOSITION: rooted in the limited scope of finite reader-equivalence; connects observational/source equivalence to a prohibition on claiming hidden identity.]

This correction absorbs the strongest probability-theory objection to the earlier definition rather than weakening the paper.

8. The Effective Agent and the Positive Separation of Encounters

The random source is only one side of the event. Define the task-relative effective agent configuration as a domain readout of retained state/history:

$$A_n^Q := q_A(S_n, T_n, c_n; Q) \quad (20)$$

[NEW DOMAIN DEFINITION: rooted in Toledo state/history plus domain translation; connects retained state and history to a task-relative effective agent without reducing the agent to propositional knowledge alone.]

Crucially, A_n^Q is not an ontology of the person. It is a partial readout chosen for a declared question. It may include conceptual organization, practical skill, embodied disposition, policy, confidence calibration, attention, salience, live accessibility, or other task-relevant retained structure. Formal incompleteness is therefore expected; the error would be to infer that an omitted distinction is causally irrelevant merely because the model omitted it.

Define the encounter itself:

$$\mathcal{I}_n^Q := \mathcal{I}_Q(\mathcal{G}_n, A_n^Q, c_n) \quad (21)$$

[NEW DOMAIN DEFINITION: rooted in Toledo domain-weld and Readout Genesis relational interpretation; connects world-side source, effective agent, and context into the task-relative agent-world encounter.]

A true "same trial" therefore requires more than source equivalence:

$$\begin{aligned} \text{SameTrial}_Q(n, m) = 1 \iff \text{SID}_Q(n, m) \in \{\text{SAME}_{\text{mech}}, \text{EQUIV}_Q\}, \\ A_n^Q \sim_Q A_m^Q, \quad \mathcal{I}_n^Q \sim_Q \mathcal{I}_m^Q. \end{aligned} \quad (22)$$

[NEW DERIVATION: rooted in Toledo finite reader-equivalence plus Eqs. 18, 20, and 21; connects world-side sameness to agent-side and interaction-side sameness.]

The constitutional non-collapse remains:

$$\text{same source readout} \not\Rightarrow \text{same effective agent} \not\Rightarrow \text{same encounter} \quad (23)$$

[NEW NON-COLLAPSE: rooted in Eq. 22; formalizes the claim that the dice may be the same while the task-relevant experiment is not.]

8.1 Interaction-Sensitivity Witness

The hostile panel correctly objected that a non-implication is too cheap. A changed agent should receive no explanatory credit unless the subsequent encounter is actually sensitive to the changed distinction. Let O_I^Q be the declared interaction reader and d_I^Q its task metric. Define:

$$\begin{aligned} \text{ISW}_Q(\mathcal{G}, A, A'; c) = \text{PASS} \iff \\ d_I^Q\left(O_I^Q \mathcal{I}_Q(\mathcal{G}, A, c), O_I^Q \mathcal{I}_Q(\mathcal{G}, A', c)\right) > \tau_I. \end{aligned} \quad (24)$$

[NEW DOMAIN DEFINITION: Interaction-Sensitivity Witness; rooted in Eq. 21. It connects a retained agent-side distinction to a material, preregistered difference in the next interaction instead of presuming that every internal change matters.]

8.2 Positive Source-Held Interaction-Separation Proposition

For the strongest source-held case, if

$$\begin{aligned} SID_Q(n, n+1) &= \text{SAME}_{\text{mech}}, & A_{n+1}^Q &\not\sim_Q A_n^Q, \\ ISW_Q(\mathcal{G}_n, A_n^Q, A_{n+1}^Q; c_n) &= \text{PASS}, \end{aligned} \quad (25)$$

[NEW DOMAIN DEFINITION: Positive Separation premises; rooted in source identity, effective-agent difference, and Eq. 24.]

then

$$\mathcal{I}_{n+1}^Q \not\sim_Q \mathcal{I}_n^Q \quad (26)$$

[NEW PROPOSITION: Positive Source-Held Interaction-Separation Proposition; rooted in Eqs. 21, 24, and 25. It is the positive form of the changed-agent thesis: under a held source, retained agent change becomes load-bearing only when a task reader detects interaction sensitivity.]

When source status is only EQUIV_Q rather than $\text{SAME}_{\text{mech}}$, Eq. 26 requires an additional robustness argument over the unresolved source fibre. The paper therefore does not upgrade finite source-equivalence into latent identity.

The proposition has a direct falsifier:

$$\begin{aligned} SID_Q(n, n+1) &= \text{SAME}_{\text{mech}}, & A_{n+1}^Q &\not\sim_Q A_n^Q, \\ ISW_Q(\mathcal{G}_n, A_n^Q, A_{n+1}^Q; c_n) &= \text{FAIL} \\ &\Rightarrow \text{no changed-agent credit for } \mathcal{I}^Q. \end{aligned} \quad (27)$$

[NEW PROPOSITION: Interaction-separation falsifier; rooted in Eq. 24. It prevents internal transformation from becoming an unfalsifiable explanation of external divergence.]

This is stronger than saying that learning changes the odds. It states exactly when a retained transformation is allowed to alter the causal meaning of "another try," and exactly when it is not.

9. From Returned Consequence to Revision, Learning, and Improvement

The retry loop respects Toledo's constitutional ordering rather than treating world feedback as something retained directly:

$$Y_{n+1}^{\text{world}} \rightarrow O_Q \rightarrow \mu_{n+1}^Q \rightarrow E_{n+1}^Q \rightarrow \text{Retain}_Q \rightarrow A_{n+1}^Q \quad (28)$$

[NEW DERIVATION: effort-domain specialization of Toledo constitutional ordering; connects world consequence to readout, meaning/experience, retention, and the next effective agent.]

Here μ^Q and E^Q are task-relative meaning and experience readouts. They are not asserted as root primitives.

A retained difference is not yet certified learning. Define a revision-bearing retry:

$$RB_Q(n) = 1 \iff A_{n+1}^Q \not\sim_Q A_n^Q \quad (29)$$

[NEW DOMAIN DEFINITION: rooted in Eq. 28; connects retained task-relevant change to retry without yet calling that change learning or improvement.]

Let $\Lambda_Q(n)$ be a preregistered learning witness — for example warranted correction, delayed transfer, calibration improvement, retained skill acquisition, or another domain-valid criterion. Certified learning requires both retained revision and a passed witness:

$$LC_Q(n) = \text{PASS} \iff RB_Q(n) = 1 \wedge \Lambda_Q(n) = \text{PASS} \quad (30)$$

[NEW DOMAIN DEFINITION: Learning Certification Gate; rooted in Toledo Exposure $\Rightarrow$ Retention $\Rightarrow$ Improvement and Eq. 29; connects retained change to domain-warranted learning evidence.]

Learning is still not improvement. Let $\Gamma_Q(n)$ be a preregistered task-improvement witness evaluated at an appropriate horizon. Define:

$$IC_Q(n) = \text{PASS} \iff LC_Q(n) = \text{PASS} \wedge \Gamma_Q(n) > \tau_G \quad (31)$$

[NEW DOMAIN DEFINITION: Improvement Certification Gate; rooted in Toledo's Exposure/Retention/Improvement non-collapse. It connects certified learning to a separately witnessed task improvement rather than treating learning as automatically beneficial.]

The architecture therefore keeps four non-equivalent states visible:

$$\text{Exposure} \neq \text{Retained Revision} \neq \text{Certified Learning} \neq \text{Improvement} \quad (32)$$

[NEW NON-COLLAPSE: rooted in Toledo A.5/H.09.v1 and Eqs. 29–31; expands the original non-collapse without replacing it.]

This matters empirically. Misinformation can alter belief; trauma can alter action; rumination can alter salience; a policy can change randomly. Those are revisions. They become learning or improvement only when the declared witnesses pass.

10. Effort Relevance: Sensitivity Before Causality, Identification Before Credit

Let $\mathcal{U}_D^{\text{adm}}(s, c, T)$ denote admissible focal controls under the declared domain, and write $R_D(s, u) := O_D(F_D(s, u, c, T))$ for the task-relative readout response at held c, T . A strengthened effort-sensitivity gate is:

$$\begin{aligned} \text{ERG}_D = \text{PASS} \iff \exists u_a, u_b \in \mathcal{U}_D^{\text{adm}} : \\ d_Q(R_D(s, u_a), R_D(s, u_b)) > \tau_E \end{aligned} \quad (33)$$

[NEW DERIVATION: strengthened Effort-Relevance Gate; rooted in Toledo stepper/domain-weld/reader-weld. It connects admissible variation in focal control to a task-relevant readout difference above a declared materiality threshold.]

Yet sensitivity is not causal identification. Let $ID_Q(u \rightarrow Y)$ denote a preregistered identification design or mechanism-specific falsifier capable of separating the focal effect from load-bearing confounding at the declared tier. Then:

$$\begin{aligned} CER_Q = \text{PASS} &\iff \\ ERG_D = \text{PASS} \wedge ID_Q(u \rightarrow Y) = \text{PASS} \end{aligned} \quad (34)$$

[NEW DOMAIN DEFINITION: Causal Effort-Relevance Certificate; rooted mathematically in Eq. 33, with epistemic admissibility interpreted through Readout Universe and the programme's falsifier discipline.]

Thus a changed outcome after more effort cannot be credited to effort merely because effort preceded the outcome. The theory requires an admissible causal route or remains at a weaker tier. It also acts one step before identification: the treatment/estimand called "effort" must itself be a warranted construct. Contemporary causal-methods work explicitly distinguishes underlying constructs from their error-prone measured versions and warns that measurement error can distort identification and interpretation [16]. A perfectly estimated effect of "attempt count" is not automatically an effect of retained learning, embodied skill, or adaptive effort.

11. Mathematical Adapters: Absorb the Theorems, Do Not Surrender the Field

Once the event has passed the relevant attribution gates, established mathematics becomes an internal adapter of the architecture rather than a rival universal explanation.

Table 1: Regime-specific mathematical routing. External theories remain external mathematics; the architecture decides when their assumptions match the typed event.

Type	Primary adapter	What it can legitimately constrain
E_0	Martingale / renewal / null-effort process	Whether persistence or stopping alone changes expected payoff under the stated stochastic assumptions.
E_1	Controlled stochastic process / bandit / MDP	How action or policy changes reward/transition distributions; exploration versus exploitation.
E_2	Partial observability / incomplete information	Decision under hidden external state or another controller not directly observed.
E_3	Stochastic or dynamic game	Mutual strategic adaptation when actions of multiple agents alter future states.
E_4	Mixed model / model averaging / structural decomposition	Joint contribution under an explicitly identified mixture.
HOLD	No licensed attribution	Prevents theorem shopping when mechanism class itself is unresolved.

11.1 Martingale and optional stopping: a null regime, not a universal defeat

Doob-style optional-stopping results show that under their conditions, stopping a martingale does not manufacture favorable expectation merely by selecting a stopping time [9]. The present theory absorbs that result completely. In an E_0 regime where focal effort does not alter the load-bearing process, "keep trying because effort itself creates edge" is rejected.

But three further inferences require separate proof. First, a payoff martingale need not imply that the agent's internal epistemic state is a martingale. Second, if retained transformation changes the transition or reward law, the later process may no longer satisfy the original martingale assumptions. Third, enlarging the filtration to "include the history" does not by itself prove that the filtration preserves the task-relevant epistemic, embodied, or phenomenological witness. Optional stopping therefore defeats naive gambler-style persistence; it defeats retained agent transformation only when the witness-preserving bridge in Eq. 14 passes.

11.2 Bandits: action value and information value are already distinct

Bandit theory explicitly treats sequential allocation under uncertainty and the trade-off between learning which action is better and collecting reward [10, 11]. This supports rather than threatens the paper's distinction:

$$V_n^{\text{learn}}(u) \not\equiv V_n^{\text{act}}(u) \quad (35)$$

[NEW NON-COLLAPSE: rooted internally in Exposure $\neq$ Retention $\neq$ Improvement and sharpened by the external bandit comparison; separates information/learning value from instrumental action value so that "no new learning" need not imply "stop."]

The paper's additional question occurs one layer earlier: what licenses an arm/reward process as the right description of the event, and what happens when the effective agent itself is transformed by the returned consequence? Predictive sufficiency for one readout is not automatically policy sufficiency for another. Two models can be observationally equivalent for short-horizon reward prediction yet imply different interventions if one represents hidden other-control and the other absorbs it into noise.

11.3 POMDPs: hidden state is not automatically lived state

POMDPs provide a principled method for planning with hidden state through belief-state updating [12]. They are a natural adapter for E_2 when a relevant environment or other-agent state is hidden. Yet a POMDP belief state is a model-dependent sufficient statistic under its assumptions. It does not automatically exhaust embodied habit, meaning, salience, or phenomenological accessibility. A claim that the belief state is sufficient for the effort question therefore requires the same domain/readout bridge as any other reduction.

11.4 Stochastic games: the right home for mutual adaptation

When both focal and external agents update, Shapley-style stochastic-game mathematics supplies an established formal home for state transitions shaped by multiple players [13]. The revised theory therefore no longer tries to force E_3 into a single-agent stochastic kernel. It routes it explicitly. What remains distinctive is the pre-game epistemic task: deciding whether the external actor is actually a strategic controller, a stochastic environment, or still unresolved.

12. Cost, Risk, and Viability: Effort Is Not an Infinite Resource

The anchor claim never required infinite physiological energy. "Open internal iteration" means only that internal revision is not forced one-to-one by external encounter count. Real continuation remains bounded by typed costs. Define a cost vector rather than a scalar sum:

$$\mathbf{C}_n(u) = (C_n^{int}(u), C_n^{ext}(u), C_n^{opp}(u), C_n^{risk}(u)) \quad (36)$$

[NEW DOMAIN DEFINITION: rooted in the finite-budget lineage retained from v0.1; connects internal burden, external expenditure, opportunity cost, and risk without assuming commensurability.]

A scalar total is not permitted unless a domain-specific translation justifies one. Internal and external costs may also couple: fatigue can change external performance, while institutional delay can become internal burden. The vector therefore describes typed cost surfaces, not independent boxes.

Let V_D be a declared viable region and let $\tau_{V_D^c}$ be the first exit time from it under the chosen local stochastic adapter. For horizon H and risk tolerance α :

$$\mathcal{U}_{n+1}^{safe} := \{u \in \mathcal{U}_D^{adm} : \mathbf{C}_n(u) \preceq \mathbf{B}_n, \mathbb{P}(\tau_{V_D^c} \leq H \mid \mathcal{F}_n, u) \leq \alpha\} \quad (37)$$

[NEW DERIVATION: risk-aware continuation set; rooted in the Toledo stepper plus finite-budget/viability lineage and explicitly adapted to stochastic settings after the v0.1 one-step deterministic critique.]

The notation $\preceq$ is componentwise unless a domain declares another admissible order. This avoids smuggling a tiny probability of catastrophic irreversible loss into an OR-rule that would continue merely because learning value is high.

13. Continue, Switch, Stop, or HOLD

Version 0.1 proposed a universal Continue/Switch/Stop threshold rule. That was too close to an unearned optimal-stopping theorem. Version 0.3 retains the decision move to the regime-specific mathematical adapter while preserving global readout, cost, and viability constraints.

Define the router:

$$\text{Route}_Q(e_n) = \begin{cases} \mathcal{A}_M, & ECT_Q(e_n) = E_0, \\ \mathcal{A}_B, & ECT_Q(e_n) = E_1, \\ \mathcal{A}_P, & ECT_Q(e_n) = E_2, \\ \mathcal{A}_G, & ECT_Q(e_n) = E_3, \\ \mathcal{A}_X, & ECT_Q(e_n) = E_4, \\ \text{HOLD}, & ECT_Q(e_n) = \text{HOLD}. \end{cases} \quad (38)$$

[NEW SYNTHESIS: regime router; rooted in the Event-Control Type and the Toledo domain-weld principle. The adapters are external mathematics; the routing architecture is new.]

Let $\mathcal{D}_n^*$ denote the decision returned by the selected local adapter from $\mathcal{F}_n$, $\mathcal{U}_{n+1}^{safe}$, V^{learn} , and V^{act} . The global decision interface is then

$$D_{n+1} = \begin{cases} \text{HOLD}, & \text{Route}_Q(e_n) = \text{HOLD}, \\ \text{STOP}, & \mathcal{U}_{n+1}^{safe} = \emptyset, \\ \mathcal{D}_n^*, & \text{otherwise.} \end{cases} \quad (39)$$

[NEW SYNTHESIS: rooted in Eqs. 35, 37, and 38; connects epistemic typing to adapter-specific continuation, switching, or stopping without pretending that one heuristic is universally optimal.]

A necessary global stop condition therefore survives:

$$\mathcal{U}_{n+1}^{safe} = \emptyset \Rightarrow \text{STOP} \quad (40)$$

[NEW PROPOSITION: sufficient, not necessary, stop condition; rooted in Eq. 37.]

Other stop conditions are delegated to the appropriate local mathematics. This is not a retreat from rational stopping. It is a stronger type discipline: a bandit, a POMDP, a stochastic game, and a one-shot safety problem should not be forced into one universal stopping equation.

14. Comparable Agents and the Stronger Effort Thesis

The original motivating problem can now be stated without giving effort credit for chance and without allowing chance to erase learning.

Consider two agents A and B facing load-bearing external conditions that are matched to tolerance ε_C under the declared reader and comparable relevant baselines. Suppose also that:

1. the event is typed E_1 , E_3 , or an identified E_4 component in which focal control is relevant;
2. $CSML_Q = \text{PASS}$ for any conditional stochastic adapter being used, and $SAL_Q^{strong} = \text{PASS}$ only when the paper makes a strong stochastic-attribution claim;
3. $ERG_D = \text{PASS}$, and causal claims are only made at the tier licensed by CER_Q ;
4. one trajectory contains retained revision and, where claimed, passes LC_Q ;

5. the effective agent states diverge and the relevant Interaction-Sensitivity Witness passes;
6. source/other-agent changes are either matched, modeled, or left unresolved rather than attributed to effort;
7. both trajectories remain inside the declared cost/risk/viability envelope.

Then adaptive internal transformation is a licensed candidate explanation of later trajectory divergence. As relevant external differences are reduced, the explanatory load carried by verified internal differences can increase; it does not become logically exclusive because stochastic residuals and hidden controls may remain.

Strong bounded thesis. *Effort matters not merely because an agent can repeat an action, but because returned consequences can transform the effective agent that meets the next event. A sequence of externally similar draws therefore need not be a sequence of equivalent experiments.*

Yet changed-agent credit is not automatic: under a held source, the positive Interaction-Separation Condition must pass; under a merely reader-equivalent source, robustness over the unresolved source fibre is required. As relevant external differences are reduced, verified agent-side differences can carry increasing explanatory load only to the extent that they survive these interaction and causal gates.

This formulation does not lower the original claim. It moves it from "effort increases success" to the more fundamental claim that *the identity and causal meaning of a retry are jointly constituted by source, reader, retained agent state, control locus, and viable action*. Probability of success is one readout of that structure, not the whole structure.

15. Adversarial Comparison: What Each Rival Formalism Still Cannot Do Alone

15.1 Probability theory

Probability theory can define random variables, filtrations, conditional laws, martingales, stopping times, and distributions with extraordinary precision. The present paper does not ask probability theory to supply an ontology it does not claim to supply. Instead it distinguishes conditional use from strong attribution: Eq. 10 allows rigorous theorem use under declared stochastic assumptions, while Eq. 11 governs the stronger move of treating stochasticity itself as load-bearing. The attack-first question is therefore not "is probability wrong?" but "which claim has its representation earned the right to constrain?".

15.2 Causal inference

Causal inference can attack the central attribution problem: effort and outcome may share hidden causes [15]. The revised CER_Q therefore requires identification rather than

temporal precedence. But causal identification also depends on variable construction and measurement. Recent methodological work emphasizes the difference between an underlying construct and its measured proxy and shows how measurement error enters causal diagrams and identification [16]. If an "effort" variable collapses retained revision, raw repetition, embodied skill, and external opportunity into one count, formal causal correctness can coexist with a conceptually impoverished estimand. The readout architecture acts before estimation by typing the object whose effect is to be estimated.

15.3 Phenomenology

Phenomenology supplies what a pure outcome process lacks: temporal thickness, embodiment, sedimentation, practical "I can," and the dependence of present experience on retained past and anticipated future [17, 18, 19]. Contemporary work continues to emphasize embodied continuity and the refinement of body schema through habituation, motor learning, and skill acquisition [20]. The paper nevertheless refuses the opposite reduction: A_n^Q is not the whole person and phenomenology does not replace stochastic mathematics. Rather, phenomenology supplies candidate load-bearing witnesses that a thin stochastic state may erase. If those witnesses are irrelevant to the query, they may be excluded; if they are load-bearing, Eq. 14 requires their preservation.

15.4 Viability and control

Viability theory asks whether trajectories can remain within admissible constraints under uncertain dynamics and provides a natural mathematical home for safe continuation [14]. It strengthens the paper's exit layer but does not identify whether observed uncertainty comes from nature, hidden other-agency, or an underspecified readout. Event typing and stochastic license still precede the viability computation.

16. The Full Attack-First Architecture

Figure 1 shows the v0.3 architecture. The key elevation is twofold: stochastic mathematics may enter conditionally without ontological overclaim, while strong stochastic attribution and cross-domain refutation require additional license; and changed-agent explanations must pass a positive interaction-sensitivity test rather than survive merely by non-collapse.

The architecture can be summarized as:

Event $\rightarrow$ Readout $\rightarrow$ Conditional/Strong Stochastic License
 $\rightarrow$ Event-Control Type $\rightarrow$ Witness-Preserving Bridge
 $\rightarrow$ Retention/Learning $\rightarrow$ Changed Agent $\rightarrow$ Interaction Sensitivity
 $\rightarrow$ Mathematical Adapter $\rightarrow$ Cost/Risk/Viability $\rightarrow$ {*CONTINUE*, *SWITCH*, *STOP*, *HOLD*} (41)

[NEW SYNTHESIS: v0.3 principal synthesis; anchored in v0.1 but adds stochastic-attribution and cross-domain-license layers before the previous retry/cost/stop architecture.]

Figure 1: Attack-first effort architecture, v0.3.

Stage	Content
1	World event / returned consequence $\rightarrow$ finite readout O_Q
2	Conditional / Strong Stochastic License
3	Event-Control Type
4	Cross-Domain Objection License $\rightarrow$ regime-specific mathematics (martingale / bandit / POMDP / game)
5	Meaning / experience under declared reader $\rightarrow$ retention / revision
6	Learning certificate $\rightarrow$ changed effective agent A_{n+1}^Q
7	Interaction Sensitivity $\rightarrow$ next agent-world encounter $\mathcal{I}_{n+1}^Q$
8	Typed cost + opportunity + risk $\rightarrow$ viable/safe continuation? $\rightarrow$ CONTINUE / SWITCH / STOP / HOLD

Conditional stochastic modeling is separated from strong stochastic attribution; external theorems cross a witness-preserving bridge; retained agent change must pass an interaction-sensitivity witness before it receives explanatory credit.

17. Empirical Programme and Falsifiers

A theory this broad must earn its domain applications separately. For each empirical implementation, preregister:

- focal agent, external controllers, and outcome reader;
- the event-control type and evidence that licenses it;
- whether stochasticity is used conditionally under $CSML_Q$ or asserted strongly under SAL_Q^{strong} ;
- the effective-agent readout, its explicit incompleteness boundary, and which retained distinctions count as task-relevant;
- the Interaction-Sensitivity Witness and threshold τ_I ;
- the learning witness Λ_Q ;
- effort sensitivity and, where claimed, causal identification design;
- internal, external, opportunity, and risk costs;
- safe/viable region, horizon, and acceptable tail risk;
- the local mathematical adapter and its assumptions;
- falsifiers for source stationarity, agent transformation, and effort contribution.

The architecture is empirically weakened if, after controlling the external regime appropriately, verified retained agent changes add no predictive or causal information about subsequent interaction. The changed-agent explanation is specifically falsified for a claimed interaction when Eq. 27 holds. Cross-domain refutation fails when the declared

bridge does not preserve a load-bearing witness, and a learning claim fails when retained revision cannot pass its preregistered learning witness.

It also accepts negative results. If E_0 is licensed and ERG_D fails, the correct result is that effort is causally null for the relevant outcome even though repeated exposure may still have action value. If another agent fully controls the relevant transition, the correct result may be that focal effort can only alter entry conditions, not the final selection. If the source class remains unresolved, the correct result is HOLD.

18. Discussion: The Field Has Moved

The v0.1 paper asked: *when does effort matter under stochastic and other-controlled outcomes?* v0.2 moved the field to *when is another try actually the same experiment?* Version 0.3 adds the positive burden: *what must be witnessed before a changed agent is allowed to make the next experiment non-equivalent?*

That shift absorbs the strongest objections rather than yielding to them.

Optional stopping becomes a theorem for a specific null-effort regime. Bandit theory becomes an adapter for skill-sensitive stochastic opportunities. POMDPs become an adapter for hidden-state decision problems. Stochastic games become the home of mutual adaptation. Phenomenology becomes an indispensable constraint on claims that the later encounter is experientially identical merely because the external generator is stationary. Causal inference becomes the gate that prevents outcome-after-effort from being misread as outcome-because-of-effort. Viability theory becomes the mathematical constraint that stops persistence from being moralized into infinite endurance.

None of these disciplines is diminished. Their local strength is preserved. What changes is the architecture of precedence: *before a formalism answers the problem, the problem must be translated into a state/readout structure that shows the formalism is addressing the load-bearing distinction.*

This produces a stronger view of effort. Effort is not a scalar quantity stored inside a person. It is a sequence of agent-world interventions whose causal meaning depends on event type, returned readout, retained transformation, source identity status, other-agent control, and viable continuation. A retry may be:

- raw repetition with no learning;
- rational opportunity exposure despite no new learning;
- retained but mistaken revision;
- certified learning that leaves policy unchanged;
- skill-bearing adaptation that changes the odds;
- a changed encounter under the same external source;
- an interaction with a strategically adapting other agent;
- or an epistemically unresolved event that should not yet be called random.

The ordinary phrase "try again" therefore hides several mathematically and phenomenologically different operations.

19. Conclusion

Randomness is not the enemy of effort theory. A declared stochastic model is a legitimate conditional regime; strong stochastic attribution is a stronger epistemic act. The category error occurs when one is silently promoted into the other, or when a theorem about a preserved payoff coordinate is universalized into a theorem about an unpreserved agent-world transformation.

The central result is not that persistence defeats chance. It is that chance, skill, other-control, learning, and lived transformation must be typed before causal credit or causal nullity is assigned. The same world-side source can remain fixed to the strongest resolution the available readout can certify while the effective agent changes. When that agent-side change passes the Interaction-Sensitivity Witness under a held or appropriately robust source comparison, the next encounter is not equivalent to the previous one even if the external generator is. When the witness fails, the theory itself withholds changed-agent credit.

The paper's strongest statement is therefore:

same external source $\nrightarrow$ same effective agent $\nrightarrow$ same encounter $\nrightarrow$ same causal meaning of retry

A complete theory of effort must explain not only whether another draw can succeed, but also what warrants treating the event as a draw, what the agent reads from it, what is retained, whether the agent is changed, whether another controller is acting, which mathematics is admissible, what another attempt is worth, and when finite cost, risk, or viability makes exit the rational result.

Appendix A. Equation Provenance and Derivation Ledger

Equation	Status	Root / source	Connection and purpose
Eq. 1	NEW NON-COLLAPSE	Toledo domain/readout separation + Readout Universe	Separates uncertainty, stochastic mechanism, other-agent control, and unresolved residual.
Eq. 2	TOLEDO EXISTING	<code>weld/M.01.v1</code>	Retained state + control + context + ordered history $\rightarrow$ next state. Mathematical anchor.
Eq. 3	TOLEDO EXISTING	<code>weld/M.02.v1</code>	Root dynamics $\rightarrow$ admissible domain translation; prevents domain words becoming root primitives.

Equation	Status	Root / source	Connection and purpose
Eq. 4	TOLEDO EXISTING	weld/M.02.v1	Reader-preservation clause: the domain reader commutes with the q_D translation; companion to Eq. 3's dynamics-preservation clause.
Eq. 5	TOLEDO EXISTING	weld/M.03.v1	Defines finite-horizon reader-equivalence; replaces v0.1's weak equal-marginal source identity.
Eq. 6	NEW PROPOSITION	repeated admissible application of Eq. 2	Finite world-side encounters → potentially multiple internal state revisions.
Eq. 7	TOLEDO EXISTING	EQ-015/M.03.v1	Retention → structure → translation → readout → meaning/experience/claim.
Eq. 8	TOLEDO EXISTING	A.5/H.09.v1	Exposure $\neq$ retention $\neq$ improvement. Anchor for revision/learning/improvement gates.
Eq. 9	NEW DOMAIN DEFINITION	Toledo reader equivalence	Finite record → admissible mechanism fibre; blocks unique hidden-cause claims from a shared readout.
Eq. 10	NEW DOMAIN DEFINITION	Eq. 9	Declared stochastic model → legitimate conditional theorem use without asserting ontic attribution.
Eq. 11	NEW DOMAIN DEFINITION	Eq. 9	Admissible mechanism fibre → strong stochastic-attribution permission.
Eq. 12	NEW DERIVATION	Eq. 11	Live stochastic + non-stochastic rivals → HOLD.
Eq. 13	NEW NON- COLLAPSE	Eqs. 10–11	Conditional stochastic modeling $\neq$ strong stochastic attribution.
Eq. 14	NEW DERIVATION	Toledo weld/M.02.v1	External formalism → effort domain only if dynamics, output, and every load-bearing witness are preserved.
Eq. 15	NEW DOMAIN DEFINITION	Eq. 14	Witness-preserving bridge → cross-domain objection license.
Eq. 16	NEW DOMAIN DEFINITION	Toledo typing + Readout Universe	Finite evidence → $E_0/E_1/E_2/E_3/E_4$ /HOLD event-control status.
Eq. 17	TOLEDO APPLICATION	weld/M.03.v1	Applies finite reader-equivalence to world-side source comparison.
Eq. 18	NEW DOMAIN DEFINITION	finite source reader	Distinguishes mechanism identity, finite equivalence, difference, and HOLD.
Eq. 19	NEW PROPOSITION	Eq. 18	Readout-equivalent source $\Rightarrow$ latent source identity.

Equation	Status	Root / source	Connection and purpose
Eq. 20	NEW DOMAIN DEFINITION	Toledo state/history + domain weld	Retained state/history → partial task-relative effective-agent readout.
Eq. 21	NEW DOMAIN DEFINITION	Toledo domain weld + Readout Genesis relational lens	Source + effective agent + context → agent-world encounter.
Eq. 22	NEW DERIVATION	source/agent/interaction equivalence	States what must remain equivalent before two retries are called the same trial.
Eq. 23	NEW NON-COLLAPSE	Eq. 22	Same source readout ⇒ same effective agent ⇒ same encounter.
Eq. 24	NEW DOMAIN DEFINITION	Eq. 21	Agent-side difference → material interaction difference under preregistered reader/threshold.
Eq. 25	NEW DOMAIN DEFINITION	SID + agent change + ISW	Positive premises required before changed-agent credit is allowed.
Eq. 26	NEW PROPOSITION	Eqs. 21, 24, 25	Positive source-held interaction separation. Central v0.3 upgrade.
Eq. 27	NEW PROPOSITION	Eq. 24	Changed agent + held source + failed sensitivity → no changed-agent credit for the claimed interaction.
Eq. 28	NEW DERIVATION	Toledo constitutional ordering	World consequence → readout → meaning/experience → retention → next effective agent.
Eq. 29	NEW DOMAIN DEFINITION	Eq. 28	Retained task-relevant change → revision-bearing retry.
Eq. 30	NEW DOMAIN DEFINITION	Toledo A.5/H.09.v1	Retained revision + learning witness → certified learning.
Eq. 31	NEW DOMAIN DEFINITION	learning gate + improvement witness	Certified learning + task gain → certified improvement.
Eq. 32	NEW NON-COLLAPSE	Toledo A.5/H.09.v1	Exposure ≠ revision ≠ learning ≠ improvement.
Eq. 33	NEW DERIVATION	Toledo stepper/domain/readout weld	Admissible effort variation → material task-readout sensitivity.
Eq. 34	NEW DOMAIN DEFINITION	ERG + identification witness	Sensitivity + causal identification → causal effort-relevance certificate.
Eq. 35	NEW NON-COLLAPSE	Toledo learning non-collapse + bandit comparison	Learning value ≠ instrumental action value.

Equation	Status	Root / source	Connection and purpose
Eq. 36	NEW DOMAIN DEFINITION	v0.1 cost anchor + viability lineage	Internal/external/opportunity/risk cost kept typed rather than illicitly summed.
Eq. 37	NEW DERIVATION	Toledo stepper + viability/budget lineage	Typed cost + stochastic exit-risk → safe continuation set.
Eq. 38	NEW SYNTHESIS	Event-Control Type + Toledo domain routing	Event regime → external local mathematical adapter.
Eq. 39	NEW SYNTHESIS	safe set + router + local adapter	Global interface for CONTINUE/SWITCH/STOP/HOLD without claiming one universal Bellman equation.
Eq. 40	NEW PROPOSITION	Eq. 37	Empty safe continuation set ⇒ STOP; sufficient, not necessary.
Eq. 41	NEW SYNTHESIS	all above	Principal v0.3 architecture: licensing → transformation → positive interaction separation → local mathematics → viable exit.

Appendix B. Attack Matrix: How the Main Reviewer Camps Are Absorbed

Reviewer camp	Attack on v0.1	v0.3 absorption	What the rival must now establish to refute the stronger claim
Martingale / optional stopping	More trials or stopping choice need not create expected edge in a fair process; an enlarged filtration can absorb more history.	Accepted completely as the E_0 null-effort adapter. CSML permits the theorem conditionally; witness-preserving CDOL blocks "bigger filtration" from being treated as automatic epistemic or phenomenological sufficiency.	Show that the stochastic representation preserves every load-bearing witness of the claim being challenged, not merely the payoff coordinate.
Probability / randomness	Equal marginals do not imply the same stochastic process; a useful stochastic model does not itself prove ontic randomness.	v0.1 equal-law identity removed; CSML is separated from strong stochastic attribution; source sameness uses finite reader-equivalence plus SAME/EQUIV/DIFF/HOLD.	A conditional theorem may stand under CSML. To make a strong randomness claim, establish SAL_Q^{strong} or state the mechanism as unresolved.

Reviewer camp	Attack on v0.1	v0.3 absorption	What the rival must now establish to refute the stronger claim
Causal inference	Output sensitivity is not causal identification; effort can be confounded by skill/resources/support.	ERG and CER split sensitivity from causal certification.	Supply an identification design or restrict the claim to sensitivity/association tier.
POMDP	Hidden state can be represented through belief states.	Accepted as E_2 adapter.	Show that the belief-state reduction preserves the load-bearing human readout/experience if used to erase phenomenological transformation.
Stochastic games	Adaptive others make a single-agent kernel inadequate.	E_3 is routed to stochastic/dynamic games rather than forced into Eq. (8) of v0.1.	Establish the player/state/action model and its bridge to the claimed social/epistemic interaction.
Phenomenology	World feedback is not retained directly; experience is temporally thick, embodied, and historically structured.	Toledo constitutional ordering moved into the mathematical backbone; effective agent broadened beyond propositional knowledge.	A reduction to a stochastic state must show that it preserves the relevant lived/embodied distinctions.
Learning theory	State change can be error, rumination, conditioning, or noise rather than learning; learning itself need not improve the task.	Revision-bearing retry, certified learning, and certified improvement are now separate gates.	Demonstrate the declared learning and improvement witnesses; otherwise retain the weaker label actually earned.
Changed-agent critique	"Same source $\Rightarrow$ same encounter" is only a weak non-implication; internal change may be irrelevant to the next interaction.	v0.3 adds the Interaction-Sensitivity Witness, a positive source-held separation proposition, and an explicit failed-witness falsifier.	Hold the source at the declared level and show that the preregistered interaction reader is insensitive to the retained agent difference.

Reviewer camp	Attack on v0.1	v0.3 absorption	What the rival must now establish to refute the stronger claim
Viability / risk	One-step feasibility and scalar cost ignore path risk, irreversibility, and tail loss.	Typed four-component cost vector plus stochastic horizon-exit risk constraint.	Critique the declared risk/viability model rather than the existence of a universal OR-threshold that v0.3 no longer claims.
Optimal stopping / DP	v0.1 Continue/Switch/Stop heuristic was not an optimal-stopping theorem.	Global heuristic replaced by a router to local optimal-control/stopping mathematics; only empty safe-set $\Rightarrow$ STOP is global.	Show routing or adapter assumptions are wrong, not merely that a different domain has a different Bellman equation.

Appendix C. Internal Literature Lineage

C.1 Toledo: mathematics and admissibility

The current Toledo registry is the sole mathematical authority for equations claimed as pre-existing. The v0.3 manuscript relies directly on `weld/M.01.v1` (state stepper), `weld/M.02.v1` (domain/reader weld), `weld/M.03.v1` (finite reader-equivalence), `EQ-015/M.03.v1` (constitutional ordering), and `A.5/H.09.v1` (Exposure $\Rightarrow$ Retention $\Rightarrow$ Improvement). All other displayed equations in this paper are explicitly marked as new derivations, definitions, propositions, non-collapses, or syntheses.

C.2 Readout Genesis: ontology

Readout Genesis governs the interpretation of retained distinction, state, translation, reader, and domain. Its central discipline is that a domain is a translation of retained structure rather than a new root object. The effort paper therefore does not install "effort," "randomness," "knowledge," or "experience" as root primitives.

C.3 Readout Universe: epistemology

Readout Universe supplies the readout-not-truth stance and tier discipline used here to distinguish mechanism identity, reader-equivalence, conditional modeling, and unresolved states. In particular, a finite record is not promoted into a unique hidden cause without a licensed inference.

C.4 Human learning, Choice, Potential, and DCP

The earlier programme literature contributes the changed-reader thesis, the distinction between feasible/live/chosen/enacted/observed, witnessed agency under finite horizon and budget, effort/time-matched learning controls, world-return feedback, and stop/reset

precedents. v0.3 does not copy their equations into Toledo; it uses them as internal conceptual lineage under the source-governance rule.

C.5 Internal adversarial artifacts

The programme's Shared-Readout Non-Identifiability, Zero-Readout, SkillMe, Finite Readout Acceleration, and Retained Multilevel Sturm projects independently reinforce three methodological commitments used here: do not infer a unique latent state from a shared finite record; do not collapse resolved zero with unresolved status; and turn adversarial failure into an explicit gate or HOLD rather than silently preserving the earlier claim. These are internal methodological precedents, not substitute sources for Toledo equations.

Appendix D. Scope of Contribution

The paper does not present martingales, optional stopping, bandits, Markov decision processes, POMDPs, stochastic games, causal identification, phenomenology of retention/habit, viability theory, or optimal stopping as contributions of this manuscript.

The proposed contribution is the ordering, licensing, and positive-separation architecture that asks, before those tools are invoked:

What warrants the event type? What does the declared reader distinguish? Has the effective agent changed? Does the interaction-sensitivity witness show that the change matters? Is this still the same agent-world encounter? Which external mathematics has an admissible bridge to that encounter? What can still be learned or achieved at finite cost and risk?

The distinctive claim is therefore not that the Human-AI Readout Programme contains a superior martingale, bandit algorithm, POMDP solver, or stochastic-game equilibrium. It claims a prior and interstitial architectural role: determining which distinctions a local formalism must preserve before its theorem can travel, and requiring a positive interaction-separation witness before a changed agent is allowed to alter the causal meaning of retry.

Appendix E. Source-Governance Rule

For future revisions:

1. Mathematical equations represented as pre-existing programme equations must be verified against the current `morrocwi/toledo` repository.
2. If a required equation is absent, inspect Toledo first for related notation, constraints, and derivations; only then introduce a new equation, explicitly labeled as new.
3. Ontological interpretation is governed by `morrocwi/readout_genesis`.
4. Epistemological interpretation is governed by `morrocwi/readout_universe`.

5. Human-AI collaboration claims, if later added, must be checked through `morrocwi/glosa`.
6. External mathematics may be cited as prior art, comparator, or local adapter but must not be represented as a Toledo equation.
7. A theorem's correctness in its native domain does not itself establish cross-domain applicability; the relevant witness-preserving bridge must be declared.
8. Conditional stochastic modeling and strong stochastic attribution must remain distinct; a theorem may be valid under an assumed stochastic model even when the event's latent mechanism remains unresolved.

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